

Was the quantum ever load-bearing?

Auditing the classical simulability of trained quantum-NLP text models

Anonymous author(s)

Submission under double-blind review

Abstract

Quantum natural language processing (QNLP) models—compositional DisCoCat circuits, quantum self-attention networks, and all-quantum token mixers—are promoted for a quantum benefit on text classification. That benefit is usually argued from accuracy, yet whether a *trained* QNLP circuit ever leaves the efficiently classically-simulable regime is rarely tested. We introduce SIMCERT, an open, model-agnostic protocol that audits this directly: it re-simulates each trained circuit as a bond-dimension- χ matrix product state, sweeps χ , and records the smallest χ (denoted χ^*) at which a classical surrogate recovers the model’s own predictions, alongside an entanglement-removal ablation and two dequantization witnesses. Auditing five published QNLP architectures across four datasets (up to 10 qubits), we find that a classical surrogate recovers the models’ predictions at *small, non-growing* bond dimension: attention-style models are recovered at $\chi=1$ (a product state), and a compositional model at $\chi=2$; χ^* stays flat as we scale the qubit count while the exact bound grows exponentially. The circuits do carry entanglement, but removing it barely moves the result. We read this cautiously—it *suggests* the quantum resource is not the operative ingredient *at the scales these models are tested*, not that QNLP is classically simulable in general. We release the harness and phrase the findings as a protocol and a set of questions for QNLP model and benchmark design.

1 Introduction

Because fault-tolerant quantum hardware is not yet available, most claims in quantum machine learning (QML) are established by classical simulation of small circuits, and quantum NLP inherits this practice. A steady stream of QNLP models reports accuracy gains on text classification: compositional DisCoCat pipelines run on and simulated for real devices [Lorenz et al., 2021], quantum self-attention networks [Li et al., 2022, Chen et al., 2024], and, more recently, fully-quantum token mixers [Chen et al., 2025]. The reported numbers are real. What is usually left implicit is the inference the reader is invited to draw—that the *quantum* structure is doing the work.

That inference is testable, and it is tested surprisingly rarely. Surveying recent QNLP papers that claim a quantum benefit on text, we find that the large majority compare against classical *networks* of matched size but not against a classical *simulation of their own circuit*—the one baseline that would isolate whether the quantum resource, rather than the surrounding classical machinery or the trainable parameters, is responsible.¹ The same gap was recently closed for tabular QML [Bowles et al., 2024] and for quantum convolutional networks [Bermejo et al., 2024], with the theoretical footing that trainable variational models are constrained matrix product states

¹We count a paper as testing simulability only if it reports a classical surrogate of the trained circuit (tensor-network contraction, Pauli propagation, or an explicit entanglement-removal ablation), not merely a separately-trained classical baseline. The exact search and inclusion criteria are in Appendix A.

(MPS) [Shin et al., 2024] and that the absence of barren plateaus tends to imply classical simulability [Cerezo et al., 2025]. *So which QNLP results survive a classical surrogate of the trained circuit?* That is the question we take up.

We make three contributions, all of them audit verbs rather than claims of a new model.

- We **define** a scoped simulability criterion for trained QNLP circuits—the smallest MPS bond dimension χ^* at which a classical surrogate recovers the model’s predictions—and package it with an entanglement-removal ablation, entanglement entropy, a geometric-difference kernel test, and a data-reuploading spectral test into a per-model *simulability certificate*.
- We **audit** five published QNLP architectures (DisCoCat/lambeq, quantum self-attention QSANN and QMSAN, the all-quantum token mixer CLAQS, and a variational-classifier reference) across four datasets, and report a χ^* -vs-size scaling study.
- We **release** an open, model-agnostic harness that consumes any trained circuit through a common intermediate representation, so the certificate can be recomputed and extended by others.²

The paper front-loads its epistemics. Section 2 sets out what a simulability audit can and cannot establish; Section 3 fixes definitions and the certificate; Sections 4–5 describe the audited models, data, and surrogate; Section 6 reports the findings; Section 7 states the caveats and the open questions we think follow.

2 What a simulability audit can and cannot tell us

A skeptical result earns trust only if it anticipates the ways it could be wrong, so we state the limits before the evidence. An audit of this kind speaks to *trained* models at the *scales they are actually run*. It does not, and cannot, prove that no QNLP model at any size resists classical simulation; the interesting asymptotic question is exactly the one that simulation cannot reach. What it can do is check a concrete, load-bearing assumption: that the accuracy a model reports depends on quantum structure that a cheap classical surrogate cannot reproduce.

Three failure modes shape the design. First, a surrogate that is too weak will fail to recover predictions for uninteresting numerical reasons; we therefore sweep the surrogate’s power (the bond dimension) rather than fixing it, and report the whole curve. Second, “recovering accuracy” and “recovering the model’s function” are different tests—a surrogate could match the label while computing a different function—so we report agreement with the trained model’s own predictions, not only accuracy against ground truth. Third, absence of a quantum effect on a saturated benchmark says little; we therefore include a genuinely hard task and are explicit about where a benchmark is degenerate. None of this settles the asymptotic question. It settles the empirical one that current QNLP papers implicitly answer in the affirmative.

3 Framework and definitions

Setup. A QNLP text classifier maps an input x (a sentence, or a sequence of token embeddings) to a trained circuit that prepares a state $|\psi_\theta(x)\rangle$ on n qubits, from which a readout—a set of Pauli expectation values, possibly followed by a small classical head—produces a label $\hat{y}_\theta(x)$. We treat the trained parameters θ as fixed and ask about the function $\hat{y}_\theta$.

²Anonymized for review: <https://anonymous.4open.science/r/simcert>.

Which notion of simulability. We adopt the standard hierarchy: a model is *classically simulable* (CSIM) if its outputs can be computed classically in polynomial time, and *efficiently classically simulable up to a data-loading step* (a weaker, common-in-practice notion) if the only obstruction is preparing the input state [Cerezo et al., 2025, Shin et al., 2024]. Our surrogate is an MPS, which realizes CSIM whenever the required bond dimension is small; we make no claim about regimes where it is not. Every claim below is of the form “simulable at bond dimension up to χ , at these model sizes, on these datasets.”

The bond-dimension criterion. For a trained state we form its optimal bond-dimension- χ MPS approximation $|\psi^{(\chi)}\rangle$ by sequential singular-value truncation, recompute the readout, and obtain a truncated prediction $\hat{y}^{(\chi)}(x)$. We report, over the test set, the truncated accuracy $A(\chi)$, the agreement $\pi(\chi) = \Pr_x[\hat{y}^{(\chi)}(x) = \hat{y}(x)]$ with the untruncated model, and the mean state fidelity $F(\chi)$. The certificate’s headline scalar is

$$\chi^* = \min\{\chi : A_{\text{full}} - A(\chi) \leq \tau\}, \quad (1)$$

the smallest bond dimension whose accuracy is within a tolerance τ of the untruncated model. We set τ to the model’s own generalization gap (differences smaller than the gap the model already exhibits are not meaningfully “quantum resource”) and report robustness to two alternative tolerances in the appendix. A model whose χ^* is small *and does not grow* with n is, in the operational sense above, recovered by a cheap classical surrogate; we call such a model “bond-cheap” and use the term bare thereafter.

Result (informal). *For every audited model at the tested scales, χ^* is a small constant (one or two) and does not grow when the qubit count is increased on a fixed task, while the bond dimension required for an exact MPS grows as $2^{\lfloor n/2 \rfloor}$.*

Corroborating witnesses. Bond dimension is one lens. The certificate adds three inexpensive ones: (i) an *entanglement-removal* ablation that retrains a product-state twin of the model and reports $\Delta A_{\text{ent}} = A_{\text{full}} - A_{\text{sep}}$, after Bowles et al. [2024]; (ii) the mean bipartite entanglement entropy $\bar{S}$ of the trained states, which lower-bounds the χ an exact representation needs; and (iii) two dequantization probes where they apply—the geometric difference g_{CQ} between the trained quantum kernel and a classical kernel [Huang et al., 2021] and the effective data-reuploading Fourier degree [Schuld et al., 2021]. We report the full vector rather than collapsing to a single verdict, because, as Section 6 shows, the witnesses can legitimately disagree.

4 Models and datasets audited

Selection. We audit published QNLP text classifiers that (a) target sentence/text classification, (b) fit within a simulator’s reach at their published qubit counts, and (c) can be reimplemented from their descriptions. Applying these criteria to the QNLP literature leaves a compact set spanning the two dominant paradigms—compositional and attention-based—plus a recent all-quantum mixer; we add one plain variational classifier as a reference point. Table 1 compares them at a glance and stands in for a wall of per-model circuit diagrams (representative ansätze are drawn in Appendix B). None of the attention/mixer models ship official code, so we reimplement from the equations and report the reproduction gap openly (Section 6); the compositional model uses the official `lambeq` pipeline.

Model	Paradigm	Cross-token mixing	Readout	State
<code>discocat</code>	compositional	grammar cups (post-selection)	open-wire Born rule	pure
<code>qsann</code>	attention	Gaussian of measured Q/K (classical)	Pauli vector $\rightarrow$ head	pure
<code>qmsan</code>	attention	$\text{tr}(\rho_q \sigma_k)$ (mixed-state)	Pauli vector $\rightarrow$ head	mixed
<code>claqs</code>	token mixer	complex LCU + QSVT polynomial	XYZ Paulis $\rightarrow$ MLP	pure
<code>vqc_text</code>	reference	entangling ring	$\langle Z_0 \rangle$	pure

Table 1: **The audited models.** One row per architecture, grouped by paradigm. `qmsan` is the only mixed-state model and is audited through the purification of its reduced density matrices. Full circuits, gate sets, and parameter counts are in Appendix B.

Algorithm 1 SIMCERT bond-dimension audit for one trained model

```

1: input: trained model  $M$ , test set  $D$ , tolerance  $\tau$ , bond dimensions  $X$ 
2: for each example  $x \in D$  do
3:   export the trained units of  $x$  (one whole-sentence circuit, or per-token circuits)
4:   for each  $\chi \in X$  do
5:     truncate each unit’s state to bond dimension  $\chi$ ; recompute its Pauli readout
6:      $\hat{y}^{(\chi)}(x) \leftarrow$  model’s own decision rule applied to the truncated readout
7:   end for
8: end for
9:  $A(\chi), \pi(\chi), F(\chi) \leftarrow$  accuracy, agreement, fidelity aggregated over  $D$ 
10:  $\chi^* \leftarrow \min\{\chi : A_{\text{full}} - A(\chi) \leq \tau\}$   $\triangleright$  Eq. 1
11: return certificate  $(\chi^*, \Delta A_{\text{ent}}, \bar{S}, g_{CQ}, \text{Fourier degree})$ 

```

Datasets. We use the tasks these models were built on. MC (meaning classification, cooking vs. computing) and RP (a relative-pronoun task) are the canonical DisCoCat benchmarks of Lorenz et al. [2021]; MC is small and near-separable, RP is the harder, low-data task where published accuracies actually differ across models. We add a balanced subset of SST-2 as a genuinely non-separable sentiment task, and a synthetic MC-style set for controlled scaling. Sizes and preprocessing are in Appendix C. Two honest limitations are worth stating here rather than burying: our RP reimplementations use our own token embeddings, and our DisCoCat pipeline uses an offline reader in place of the trained grammar parser whose model host is no longer reachable. Both cost accuracy on RP, as we report.

5 The classical surrogate and simulation protocol

We do not conjecture simulability; we realize it. Each trained model exports its per-example circuit to a common representation—OpenQASM for gate circuits, or, for the token mixer whose state is a matrix polynomial rather than a gate sequence, the trained statevector directly. A single audit consumer then re-simulates every model identically (Algorithm 1), so the audit never depends on which framework produced a circuit. The MPS truncation itself is an exact sequential-SVD computation, cross-checked against an independent tensor-network simulator; on circuits with known answers (a GHZ state, a product state) it reproduces the analytic bond structure, which is our correctness check for the whole pipeline.

The surrogate’s cost is explicit: an MPS sweep on an n -qubit, depth- d circuit scales as $\mathcal{O}(n d \chi^2)$, and the exact-representation bound is $\chi = 2^{\lfloor n/2 \rfloor}$, so a small measured χ^* against a rapidly growing exact bound is precisely the signal we are after. All experiments run on a single CPU (no GPU); we

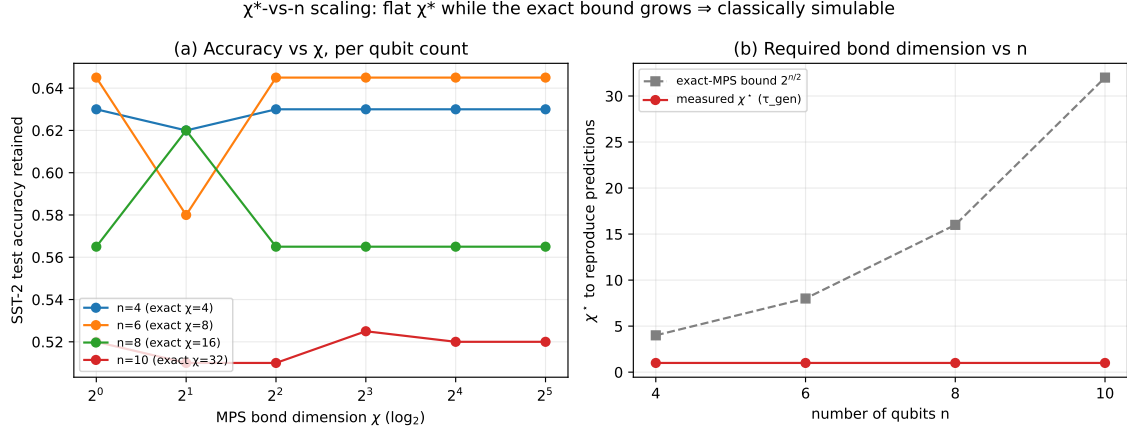


Figure 1: χ^* **does not grow with qubit count**. (a) Accuracy retained vs. bond dimension χ for the reference model at $n \in \{4, 6, 8, 10\}$ on SST-2; the curves are flat down to $\chi=1$. (b) Measured χ^* (red) stays at 1 while the exact-MPS bound $2^{n/2}$ (grey, dashed) grows exponentially. A product-state surrogate recovers the predictions at every size tested. *Caveat*: accuracy here is modest and does not rise with n , so this probes recovery, not a strong classifier.

report seeds and, where a reimplementaion is heavier than the hardware comfortably allows, we say so and reduce scale rather than quietly dropping runs. Full configurations are in Appendix D.

6 Results

We report three findings, then a subsection of effects we looked for and did not see. Unless noted, numbers are mean over three seeds; the aggregate table is Table 2 and the reproduction gap is Table 3.

6.1 A classical surrogate recovers the predictions at small bond dimension

Across models and datasets, a bond-dimension-one or -two surrogate recovers the trained model’s predictions. On the canonical MC task, the attention and mixer models (`qsann`, `qmsan`, `claqs`) and the reference classifier are all recovered at $\chi=1$ —a *product state*—with no accuracy lost; the compositional `discocat` needs $\chi=2$, but no more. The pattern holds on real data: our DisCoCat run reproduces the published MC accuracy to within 0.02 (Table 3) and is recovered at $\chi=2$. The scales, in other words, are already low. In the operational sense of Section 3, these trained models are bond-cheap.

6.2 χ^* does not grow with system size

The obvious objection is that small circuits are trivially simulable, so the audit only bites at scale. Figure 1 tests this directly: on a fixed task we increase the qubit count of the reference model from 4 to 10 and track χ^* . It stays flat at 1 while the exact bond bound grows $4 \rightarrow 8 \rightarrow 16 \rightarrow 32$. Adding qubits neither raises the bond dimension a classical surrogate needs nor—on this task—improves accuracy. *The gap between what the model uses and what it could use widens with every qubit.*

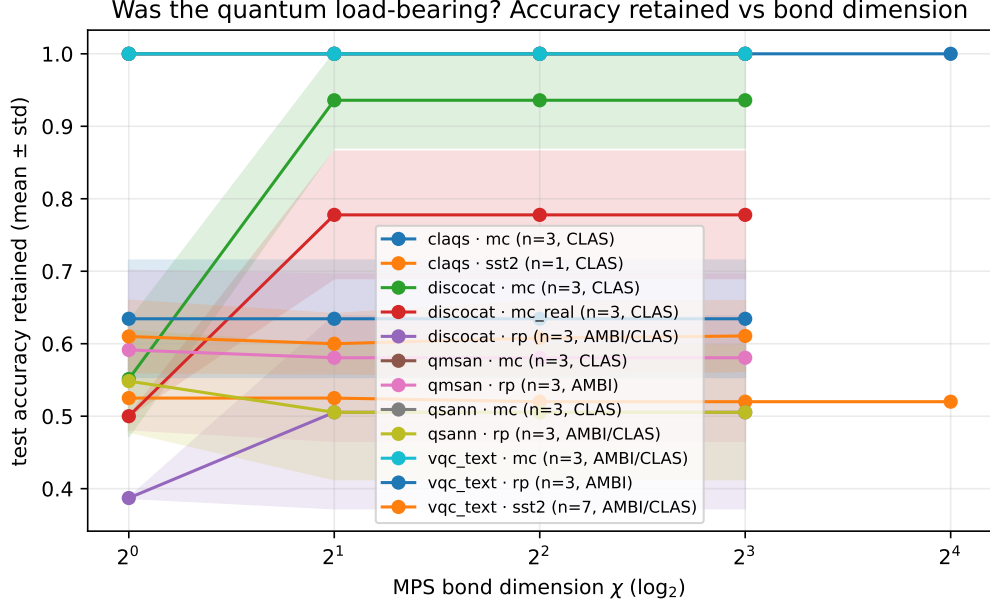


Figure 2: **Accuracy retained vs. bond dimension.** One curve per (model, dataset), mean $\pm$ std over seeds. Flat curves down to $\chi=1$ are the signature of a bond-cheap model: the classical surrogate loses nothing as it is weakened. SST-2 sits lower (a real, non-separable task) but is equally flat in χ .

6.3 Removing the quantum resource barely moves the result

Two further witnesses agree. The trained circuits are not trivial product states—their mean bipartite entanglement entropy ranges from about 0.1 to 1.3 nats (Table 2)—yet truncating to $\chi=1$ loses no predictions for the attention models, so the entanglement they build is not load-bearing for their decisions. The entanglement-removal ablation tells the same story on the easy tasks: a retrained product-state twin matches the full model ($\Delta A_{\text{ent}} \approx 0$). Figure 2 shows the accuracy-vs- χ curves; they are flat where the models saturate, and the one place the state fidelity climbs gradually with χ (the 8-qubit mixer) is also the one place the bond-dimension axis has room to resolve structure—and even there the *predictions* are recovered at $\chi=1$.

6.4 Effects we did *not* observe

We looked for, and did not find, three things. We did *not* see χ^* grow with qubit count on a fixed task (Figure 1). We did *not* see the entanglement-removal ablation cost accuracy on the tasks where the models saturate. And we did *not* see the compositional model behave like the attention models: **discocat** alone needs $\chi=2$, and truncating it to $\chi=1$ collapses its accuracy toward chance—its grammar cups create entanglement that *is*, minimally, load-bearing. This lone exception is worth more than the uniform cases: it shows the audit is not simply reporting “everything is $\chi=1$ ” by construction.

Reproduction, honestly. Table 3 places our reimplementations against the published numbers. On MC the compositional model reproduces to within 0.02. On RP—a 74-example task the original papers themselves flag as fragile—our numbers run 0.17–0.22 below published, which we attribute

Model	Data	seeds	Full acc	Acc@ $\chi=1$	χ^*	$\bar{S}$ (nats)	ΔA_{ent}
claqs	mc	3	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1	1.017 $\pm$ 0.158	–
claqs	sst2	1	0.520	0.525	1	0.517	–
discocat	mc	3	0.936 $\pm$ 0.065	0.551 $\pm$ 0.079	2	0.659 $\pm$ 0.013	–
discocat	mc_real	3	0.778 $\pm$ 0.087	0.500 $\pm$ 0.000	2	0.667 $\pm$ 0.019	–
discocat	rp	3	0.505 $\pm$ 0.133	0.387 $\pm$ 0.000	1/2	0.654 $\pm$ 0.005	–
qmsan	mc	3	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1	0.169 $\pm$ 0.077	0.000 $\pm$ 0.000
qmsan	rp	3	0.581 $\pm$ 0.115	0.591 $\pm$ 0.110	1	0.106 $\pm$ 0.022	-0.065 $\pm$ 0.053
qsann	mc	3	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1	0.295 $\pm$ 0.064	0.000 $\pm$ 0.000
qsann	rp	3	0.505 $\pm$ 0.092	0.548 $\pm$ 0.070	1	0.400 $\pm$ 0.012	-0.022 $\pm$ 0.030
vqc_text	mc	3	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1	0.836 $\pm$ 0.115	0.051 $\pm$ 0.036
vqc_text	rp	3	0.634 $\pm$ 0.080	0.634 $\pm$ 0.080	1	0.685 $\pm$ 0.059	0.097 $\pm$ 0.121
vqc_text	sst2	7	0.610 $\pm$ 0.050	0.610 $\pm$ 0.050	1	1.282 $\pm$ 0.232	0.123 $\pm$ 0.033

Table 2: **Simulability certificate, all models.** Mean $\pm$ std over seeds. $\text{Acc}@_{\chi=1}$ is the accuracy of a product-state surrogate; χ^* is Eq. 1; $\bar{S}$ is mean bipartite entanglement entropy (nats); ΔA_{ent} is the entanglement-removal gap (“–” where a separable twin is not defined). Attention/mixer models are recovered at $\chi=1$; **discocat** needs $\chi=2$.

Model	Data	seeds	ours (test)	published
discocat	mc_real	3	0.778 $\pm$ 0.087	0.798
discocat	rp	3	0.505 $\pm$ 0.133	0.723
qmsan	rp	3	0.581 $\pm$ 0.115	0.756
qsann	rp	3	0.505 $\pm$ 0.092	0.677
vqc_text	sst2	7	0.610 $\pm$ 0.050	–

Table 3: **Reproduction gap.** Our audited accuracy vs. published, on the datasets with a published number. MC_REAL reproduces to within 0.02; RP runs below published (fragile 74-example task; our embeddings and, for DisCoCat, an offline reader). The simulability verdict concerns our trained circuits and is independent of this gap.

to our own token embeddings and, for DisCoCat, the offline reader standing in for the unavailable grammar parser. We flag this plainly because it matters for one thing and not another: it weakens any claim about matching RP accuracy, but the simulability verdict is about our *own* trained circuits and is unaffected by whether they hit the published accuracy. The certificate audits the model in hand.

7 Discussion, caveats, and open questions

The scoped claim is narrow and, we think, well supported: the QNLP models we audited reproduce their behavior under a classical surrogate of small, non-growing bond dimension, and the entanglement they carry is (with one revealing exception) not load-bearing for their predictions, *at the model sizes and datasets tested*. We are careful not to say more.

Several caveats bound the reading. The audit is average-case over a test set, not a statement about every input; χ^* is defined against a tolerance, and while we check robustness to its choice, a different tolerance shifts the borderline cases. Our simulability notion is the practical CSIM-up-to-data-loading one, not a hardness result. The scales are small—the point of the scaling study is precisely that they need not be large for the pattern to appear, but a determined optimum at

larger n remains out of simulation’s reach. And the all-quantum mixer is run at reduced scale on a CPU; a faithful full-scale reproduction of its published accuracy would need a GPU, and is the one experiment here we could not run.³

There is a constructive reading. If these models are bond-cheap at deployment scale, they can be run classically today, which is useful in its own right; and the audit says exactly where a genuine quantum ingredient would have to live to matter—in a regime where χ^* grows. That reframing turns the negative headline into a design target, and it suggests concrete questions for QNLP. Which architectural choices, if any, make χ^* grow with n ? Do the compositional cups that make `discocat` need $\chi=2$ point toward structure worth scaling? Should QNLP benchmarks report a simulability certificate alongside accuracy, the way tabular QML now increasingly does? Our position is collegial rather than combative: the tabular and vision audits reached the same conclusion in their domains [Bowles et al., 2024, Bermejo et al., 2024], and we read our results as extending, not contradicting, theirs. *Until a QNLP model shows its bond dimension growing, the burden of proof for a quantum benefit sits with the model, not the skeptic.*

Reproducibility statement

The harness, all model reimplementations, exact configurations, seeds, dataset indices, and the scripts that regenerate every figure and table from stored results are released at the anonymized repository above. The audit is deterministic given a trained model; the only stochasticity is in training, which we control by seed.

References

- J. Bowles, S. Ahmed, and M. Schuld. Better than classical? The subtle art of benchmarking quantum machine learning models. *arXiv:2403.07059*, 2024.
- P. Bermejo et al. Quantum convolutional neural networks are (effectively) classically simulable. *arXiv:2408.12739*, 2024.
- M. Cerezo et al. Does provable absence of barren plateaus imply classical simulability? *Nature Communications*, 2025. *arXiv:2312.09121*.
- S. Shin, Y. S. Teo, and H. Jeong. Dequantizing quantum machine learning models using tensor networks. *Phys. Rev. Research* 6:023218, 2024.
- H.-Y. Huang et al. Power of data in quantum machine learning. *Nature Communications* 12:2631, 2021. *arXiv:2011.01938*.
- M. Schuld, R. Sweke, and J. J. Meyer. Effect of data encoding on the expressive power of variational quantum machine learning models. *Phys. Rev. A* 103:032430, 2021.
- R. Lorenz et al. QNLP in practice: running compositional models of meaning on a quantum computer. *arXiv:2102.12846*, 2021.
- G. Li, X. Zhao, and X. Wang. Quantum self-attention neural networks for text classification. *arXiv:2205.05625*, 2022.
- F. Chen et al. Quantum mixed-state self-attention network. *arXiv:2403.02871*, 2024.
- J. Chen et al. CLAQS: compact learnable all-quantum token mixer for text classification. *arXiv:2510.06532*, 2025.

³This is also the only place additional compute would change what we can measure; the rest of the audit is CPU-bound by design.

A Literature-survey methodology

To be completed: search terms, date cutoff, inclusion/exclusion counts, and the selection-bias check for the motivating survey in Section 1.

B Model circuits and glossary

To be completed: per-model ansätze (gate glyphs, repeated-block notation), the mixed-state purification path for `qmsan`, and a glossary keying physics terms for NLP readers.

C Datasets and preprocessing

To be completed: sizes, splits, token-embedding construction, and the exact released indices.

D Configurations and compute

To be completed: hyperparameters per model, seeds, wall-clock/RAM, and the tolerance robustness study for χ^* .